

REMOTE CONTROL FOR HOME APPLIANCES

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Using this circuit, you can remotely control the switch-on and switch-off operation of your AC mains operated home appliances. The working range of the circuit depends on the orientation and the intensity of the IR beam.

The circuit consists of a transmitter and a receiver.

Fig. 1 shows the transmitter circuit. It is built around timer IC NE555 wired as an astable multivibrator. The multivibrator produces a pulsed output waveform with 'on' time of about 57

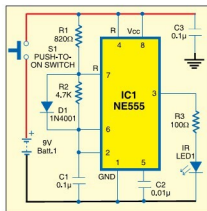


Fig. 1: Transmitter circuit

μs and 'off' time of about 326 μs, which means it generates about 2.6 kHz. The output of IC1 is fed to IR LED1 through current-limiting resistor R3. The IR LED1 used here is the same as in TV remotes. The circuit operates off a 9V battery, which is connected to the circuit through switch S1.

Fig. 2 shows the receiver circuit. It consists of phototransistor L14F1 (T1), voltage regulator 7805 (IC2), three 2N2222 transistors (T2, T3 and T4), dual voltage converter LM319 (IC3), dual J-K flip-flop 74109 (IC4) and some discrete components. The circuit operates off a 9V battery, which can be connected to the circuit through switch S2.

The Darlington pair built around transistors T2 and T3 amplifies the photo-current generated by the phototransistor (T1). The equivalent photo-voltage appears across resistor R4. So across resistor R4 you get a replica (in term of wave shape but not in amplitude) of what you produce at the output of IC1 in the transmitter. The amplitude would vary with distance and other factors such as the angle of arrival of the IR beam at sensor L14F1.

The low-pass filter constituted by resistor R7 and capacitor C4 produces

about 3V DC. This DC voltage is fed to the junction of the inverting input of N1 and the non-inverting input of N2. The window comparator (IC3) is designed such that whenever the input voltage is between 2 and 4 volts (greater than 2V but less than 4V), its output goes high. If the input voltage is less than or equal to 2V, or more than or equal to 4V, the output goes low.

The window output is fed to the clock input of J-K flip-flop CD74109 (IC4). IC4 is wired in toggle mode. That means its output goes high if it is initially low and vice versa every time it is clocked. The output of IC4 is fed to the base of relay-driver transistor T5. Relay RL1 energises to light up the bulb when transistor T5 conducts.

Working of the circuit is simple. Initially, when no IR beam is falling on sensor photo-transistor T1, the DC voltage appearing at the input of the window comparator is nearly zero. The window output remains low. Transistor T5 is cut-off and the relay remains de-energised.

When switch S1 is pressed momentarily, the IR beam falls on the phototransistor for this short period of time and a positive-going pulse appears at

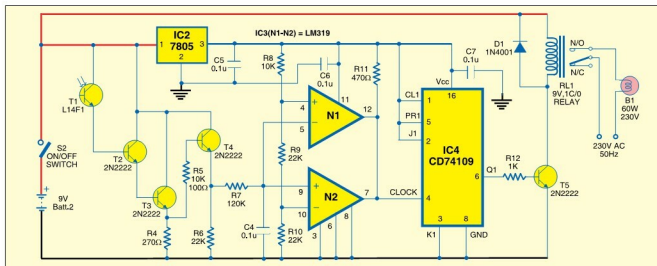


Fig. 2: Receiver circuit

